

ECO-LOOP

Autonomous LLM-Driven HVAC Control for Commercial Buildings

PROBLEM STATEMENT ID	Question 1
PROBLEM STATEMENT TITLE	Autonomous AI-Driven Dynamic Building Energy Optimization
THEME	Clean & Green Technology
PS CATEGORY	Software
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Turning Static Buildings Into Self-Correcting Physical AI

1

The Idea

A closed-loop autonomous agent that pairs NREL's EnergyPlus physics simulator with a local, open-source LLM reasoning loop (ReAct) to observe live zone conditions and continuously re-decide heating and cooling setpoints — instead of running on a fixed schedule.

2

How It Solves The Problem

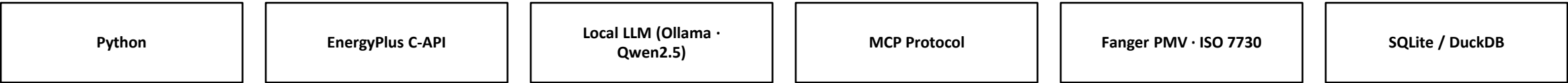
Buildings consume roughly 40% of global energy, yet most Building Management Systems still run on static, decades-old rule schedules that can't react to real-time weather, occupancy, or grid-price signals. Eco-Loop replaces that fixed schedule with a system that reasons every cycle.

3

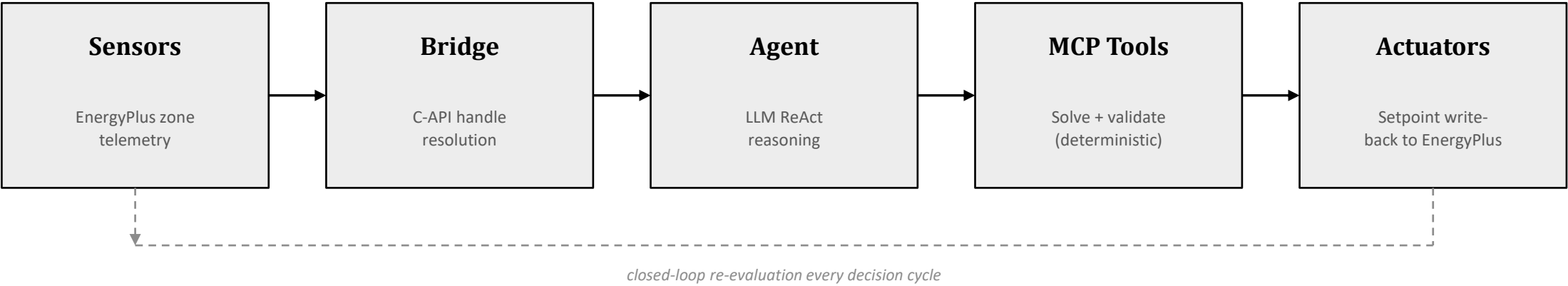
Innovation & Uniqueness

The LLM only proposes intent — it never computes setpoint math or writes to an actuator. Every decision passes through a deterministic solver and is validated twice against hard safety bounds before it reaches the building, over a standardized MCP tool-calling interface.

Technology Stack & Control Pipeline



METHODOLOGY — THE CLOSED CONTROL LOOP



Grounded in Proven Building Blocks

WHY IT'S FEASIBLE

- Built on NREL's production-grade EnergyPlus engine — no unproven physics to validate from scratch.
- Comfort model follows ISO 7730 / ASHRAE 55, established international standards, not novel unverified math.
- Fully local, open-source LLM stack (Ollama) — no cloud dependency, no per-call API cost, data stays on-site.
- Modular, single-host architecture — deployable as a working proof-of-concept within a short PoC timeline.

RISKS & MITIGATION STRATEGY

RISK

Full-year run too slow given LLM latency

→ Representative-day sampling strategy

RISK

An unsafe setpoint slips through

→ Two independent validator checks before every actuation

RISK

Local LLM server crashes or times out

→ Automatic fallback holds last known-good setpoints

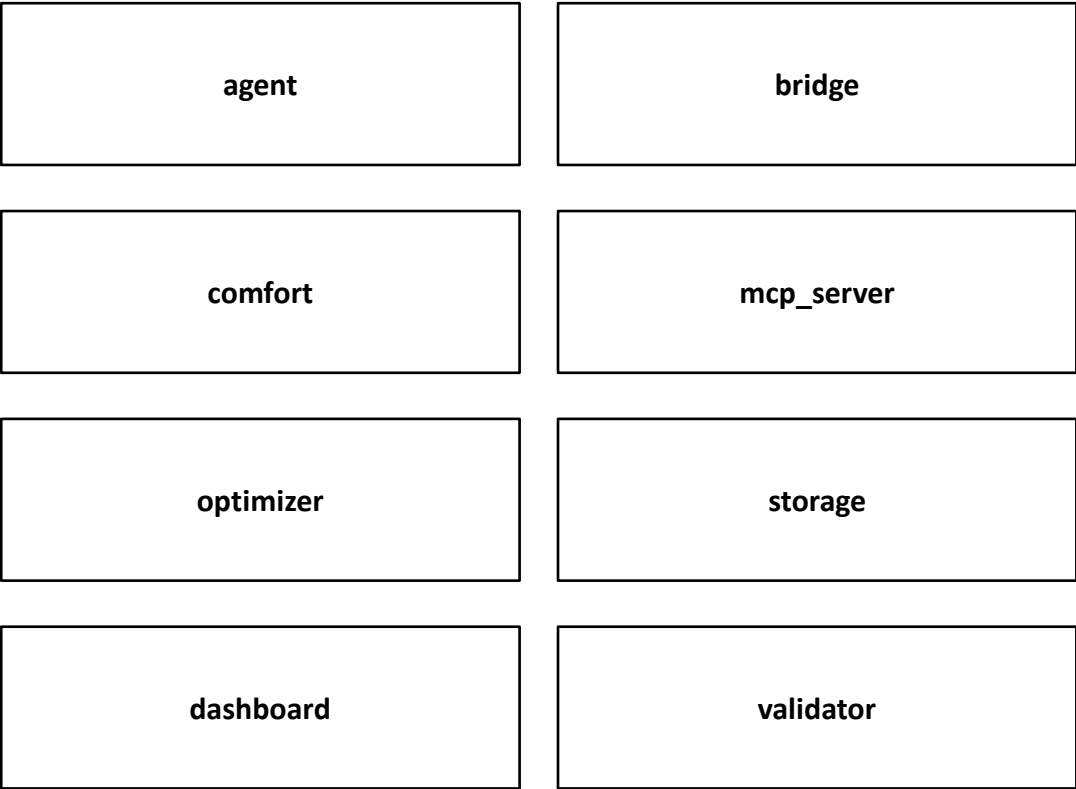
RISK

Agent context grows over multi-day runs

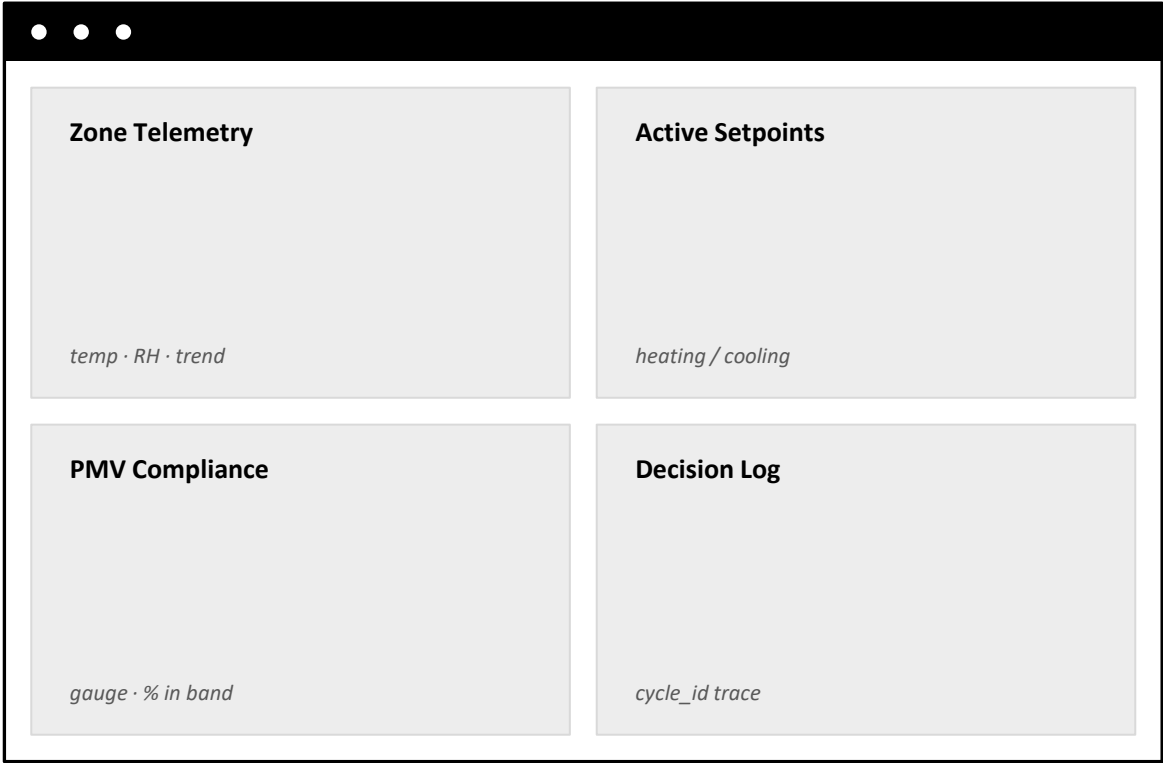
→ Two-tier memory: rolling window + periodic reflection

System Modules & Monitoring Dashboard

MODULE MAP (src/)



LIVE MONITORING DASHBOARD (read-only)



Code repository · working prototype · recorded demo available alongside this submission.

Grounded in Standards & Prior Work

SIMULATION & COMFORT STANDARDS

- NREL — EnergyPlus Runtime API Documentation
- ANSI/ASHRAE Standard 55-2023 — Thermal Environmental Conditions for Human Occupancy
- ISO 7730:2005 — Analytical Determination of PMV/PPD Thermal Comfort
- Fanger, P.O. — Thermal Comfort: Analysis and Applications (1970)

AI AGENTS & CONTROL

- Yao et al. — ReAct: Synergizing Reasoning and Acting in Language Models, ICLR 2023
- Drgoňa et al. — All You Need to Know About MPC for Buildings, Annual Reviews in Control, 2020
- Wang & Hong — RL for Building Controls: Opportunities & Challenges, Applied Energy, 2020
- Model Context Protocol Specification — Anthropic / MCP Steering Committee